

4	(a)	<i>Magnetic flux</i> of a region is defined as <u>the product of the component of the magnetic flux density that is perpendicular to the surface and the area of the surface</u>. Its unit is weber (Wb).	B1 B1
	(b)	Rate of increase of area = $(3 - (-5)) \times 0.15$ $= 1.2 \text{ m}^2 \text{ s}^{-1}$	C1 A1
	(c)	As the area of loop ABDCA increases with time, the flux linkage through the area increases. By Faraday's law, an e.m.f. induced is proportional to the rate of change of flux linkage. Since the rate of change of flux linkage increases, an e.m.f. is induced between the ends of the rods.	B1 B1
	(d)	e.m.f. induced $= \left - \frac{d\Phi}{dt} \right $ $= B \frac{dA}{dt} = (0.15)(1.2) = 0.18 \text{ V}$	C1 A1
	(e)	Because the area is increasing, the flux linkage is increasing. By Lenz's law, the loop will tend to create a magnetic field to oppose the applied magnetic field. By the right hand grip rule, the induced current will flow in the anticlockwise direction of ABDCA. OR For rod AB, by Lenz's law, the magnetic force will oppose the motion and thus point to the right. By Fleming's left hand rule, the induced current will flow from A to B. Similarly, for rod CD, a larger induced current will flow from D to C, since rod CD is moving at a greater speed. Hence, the net effect is that the induced current will flow in the anticlockwise direction of ABDCA.	B1 B1 B1 B1
	(f)	The rate of increase of area (dA/dt) is now of a smaller magnitude. Hence the induced e.m.f. will have a smaller magnitude as well. OR The induced current due to the motion of rod AB will now flow from B to A, opposing the current induced by the motion of rod CD. The currents will partially cancel, thus the induced e.m.f. will be smaller.	B1 B1 B1 B1